Thick Reality Front – exploratory numerical framework

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We introduce a computational framework for exploring a "thick reality front" description of quantum dynamics. Instead of associating physical reality with an instantaneous wavefunction at a single time, the model assumes that observable structure emerges from a finite temporal region surrounding the present. The framework combines forward quantum evolution with backward propagation conditioned on detection events, producing an effective density field that depends on both preparation and measurement. Numerical simulations of a double-slit configuration demonstrate the emergence of coherent ridge trajectories in the probability density evolution. The approach is compatible with standard quantum predictions while offering a geometric interpretation of probability flow in spacetime.

INTRODUCTION

Quantum mechanics describes the evolution of a system through the Schrödinger equation

$$i\hbar\partial_t\psi = H\psi. \tag{1}$$

While the equation predicts the evolution of the wavefunction, measurement outcomes appear to select definite results that are not described by unitary evolution alone.

Several interpretations address this issue, including collapse models, Bohmian mechanics, and time-symmetric formulations of quantum mechanics. In this work we explore a computational framework inspired by time-symmetric ideas but formulated as a numerical simulation method.

The key concept is that observable structure may not correspond to a single time slice of the wavefunction. Instead, the effective "reality" may emerge from a finite temporal region surrounding the present. We refer to this region as a *thick reality front*.

THICK REALITY FRONT FORMALISM

Forward evolution

Consider a system prepared at time t_0 with initial state ψ_0. Forward evolution is given by

$$\psi_f(t) = U(t, t_0)\psi_0. \tag{2}$$

For Schrödinger dynamics the wavefunction satisfies

$$i\hbar\partial_t\psi_f(x,t) = H\psi_f(x,t). \tag{3}$$

The forward probability density is

$$\rho_f(x,t) = |\psi_f(x,t)|^2. \tag{4}$$

Backward propagation

Suppose a detection occurs at position x_d and time t_d. We define a localized detection state

$$\phi(x, t_d) = \exp\left(-\frac{(x-x_d)^2}{2\sigma_c^2}\right). \tag{5}$$

Backward propagation is defined through

$$\phi(t) = U^\dagger(t_d, t)\phi(t_d). \tag{6}$$

Temporal thickness

Instead of associating reality with a single detection time, we consider a temporal window around t_d. Let t_k denote candidate detection times within a region of width σ_T.

The temporal weights are

$$w_k = \frac{\exp[-(t_k - t_d)^2/(2\sigma_T^2)]}{\sum_j \exp[-(t_j - t_d)^2/(2\sigma_T^2)]}. \tag{7}$$

Retrodictive field

Backward states generate a retrodictive weighting field

$$E_{\text{mix}}(x,t) = \sum_k w_k |\phi_k(x,t)|^2. \tag{8}$$

Thick reality front density

Observable structure emerges from the product of forward probability and retrodictive weighting.

$$\Gamma(x,t) = \rho_f(x,t) E_{\text{mix}}(x,t). \tag{9}$$

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To obtain a normalized distribution we define

$$\rho_{\mathrm{TRF}}(x,t) = \frac{\rho_f(x,t)E_{\mathrm{mix}}(x,t)}{\int dx\, \rho_f(x,t)E_{\mathrm{mix}}(x,t)}. \tag{10}$$

In two dimensions

$$\rho_{\mathrm{TRF}}(x,y,t) = \frac{|\psi_f(x,y,t)|^2 E_{\mathrm{mix}}(x,y,t)}{\iint dx\, dy\, |\psi_f(x,y,t)|^2 E_{\mathrm{mix}}(x,y,t)}. \tag{11}$$

RIDGE TRAJECTORIES

The density ρ_{TRF} often develops narrow high-probability structures. These can be tracked through ridge trajectories defined by

$$\mathbf{r}_R(t) = \arg\max_{\mathbf{r}} \rho_{\mathrm{TRF}}(\mathbf{r},t). \tag{12}$$

These curves provide a geometric description of probability flow through the thick reality front.

VELOCITY FIELDS AND BOHMIAN COMPARISON

The quantum probability current is

$$\mathbf{j} = \frac{\hbar}{m}\mathrm{Im}(\psi^* \nabla \psi). \tag{13}$$

This defines a velocity field

$$\mathbf{v} = \frac{\mathbf{j}}{\rho_f}. \tag{14}$$

In Bohmian mechanics these velocities define particle trajectories. In the present framework we compare these trajectories with ridge trajectories extracted from ρ_{TRF}.

CONNECTION TO THE BORN RULE

The thick reality front formalism does not modify the underlying quantum dynamics. Forward evolution remains governed by the Schrödinger equation.

Detection probabilities are determined by the forward density

$$P(x_d,t_d) \propto |\psi_f(x_d,t_d)|^2, \tag{15}$$

which is precisely the Born rule.

density_evolution.png

FIG. 1. Example density evolution in a double-slit configuration.

Backward propagation only redistributes weight inside the thick reality front but does not change the detection probability itself.

Integrating the thick-front density over intermediate times yields

$$\int \rho_{\mathrm{TRF}}(x,t)\, dx = 1, \tag{16}$$

and the marginal detection distribution remains

$$P(x_d) = |\psi_f(x_d,t_d)|^2. \tag{17}$$

Thus the framework is consistent with the standard Born probability rule.

NUMERICAL EXPERIMENT

We simulate a two-dimensional double-slit experiment using a split-operator Fourier method. A Gaussian wave packet propagates toward a barrier containing two narrow slits.

The evolution sequence is

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wave packet -> double slit -> interference -> screen
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Forward evolution produces the interference pattern while backward propagation generates the retrodictive field.

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ridge_vs_bohmian.png

FIG. 2. Comparison of ridge trajectories and Bohmian trajectories.

RELATIVISTIC EXTENSION

The framework can be extended to relativistic dynamics. For the Dirac equation

$$i\hbar\partial_t\psi = (c\boldsymbol{\alpha}\cdot\mathbf{p} + \beta mc^2)\psi, \tag{18}$$

the conserved current is

$$j^\mu = \bar{\psi}\gamma^\mu\psi. \tag{19}$$

The velocity field becomes

$$\mathbf{v} = \frac{\mathbf{j}}{j^0}. \tag{20}$$

The thick reality front construction can therefore be applied to relativistic wave dynamics.

CONCLUSION

We introduced a computational framework for exploring thick reality front models of quantum dynamics.

By combining forward evolution with backward propagation from detection events, the method produces structured probability flows that can be analyzed geometrically through ridge trajectories.

The framework reproduces standard quantum predictions while providing a new spacetime interpretation of probability flow and measurement.

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